Review Questions

1. When data cannot be changed after a class is compiled, the data is _____.

    a. variable

    b. constant

    c. volatile

    d. mutable

    Answer: b

    Feedback:

    Constant data cannot be changed after a class is compiled. Variable data can be changed after a class is compiled. Volatile describes memory content that is lost when power is lost. A state that can be changed is mutable; a state that cannot be changed is immutable.

2. Which of the following is not a primitive data type in Java?

    a. `boolean`

    b. `byte`

    c. `sector`

    d. `int`

    Answer: c

    Feedback:

    The `sector` type is not a primitive data type in Java. The `boolean` type, `byte` type, and `int` type are all primitive data types.

3. Which of the following elements is not required in a variable declaration?

    a. a type

    b. an identifier

    c. an assigned value

    d. a semicolon

    Answer: c

Feedback:

An assigned value is not required in a variable declaration but it is optional. A type, an identifier, and a semicolon are all required.

4. The assignment operator in Java is _____.

   a. `=`

   b. `==`

   c. `:=`

   d. `::`

   Answer: a

   Feedback:

   The assignment operator in Java is `=`.

5. Assuming you have declared `shoeSize` to be a variable of type `int`, which of the following is a valid assignment statement in Java?

   a. `shoeSize = 9;`

   b. `shoeSize = 9.5;`

   c. `shoeSize = "nine";`

   d. `shoeSize = 9.5F;`

   Answer: a

   Feedback:

   The value `9` can be assigned to an `int`. The value `9.5` cannot be assigned to an `int` because an `int` cannot contain a decimal. By default, `9.5` is a `double`. The value `"nine"` cannot be assigned to an `int` because it is a `String`. The value `9.5F` cannot be assigned to an `int` because an `int` cannot contain a decimal. The value `9.5F` is a `float`.

6. Which of the following data types can store the value 0 using the least amount of memory?

   a. `short`

   b. `long`

   c. `int`

   d. `byte`

Answer: d

Feedback:

A `byte` is 1 byte in size; it is the shortest of the data types. A `short` can hold a 0, but it is not the shortest data type; it occupies 2 bytes. A `long` can hold a 0, but it is not the shortest data type; it occupies 8 bytes. An `int` can hold a 0, but it is not the shortest data type; it occupies 4 bytes.

7. Correct. A `boolean` variable can hold _____.

    a. a character

    b. a whole number

    c. a decimal number

    d. the value `true` or `false`

    Answer: d

    Feedback:

    A `boolean` variable can hold `true` or `false`. A `boolean` variable cannot hold a character; the data type `char` is used for characters. A `boolean` variable cannot hold a whole number; the data types `int`, `byte`, `short`, and `long` are used to hold whole numbers. A `boolean` variable cannot hold a decimal number; the data types `double` and `float` are `used` to hold decimal numbers.

8. The value 137.68 can be held by a variable of type _____.

    a. `int`

    b. `short`

    c. `double`

    d. `byte`

    Answer: c

    Feedback:

    137.68 can be stored in a `double`. 137.68 cannot be stored in an `int`, `short`, or `byte`. Those types cannot accommodate decimal places.

9. An escape sequence always begins with a(n) _____.

    a. `e`

    b. forward slash

c. backslash

d. equal sign

Answer: c

Feedback:

An escape sequence begins with a backslash.

10. Which Java statement produces *w* on one line and *xyz* on the next line?

a. `System.out.println("wxyz");`

b. `System.out.println("w " + "xyz");`

c. `System.out.println("w\nxyz");`

d. `System.out.println("w\nx\ny\nz");`

Answer: c

Feedback:

Printing `"w\nxyz"` inserts a new line after `w`. Printing `"wxyz"` or `"w " + "xyz"` places all the characters on the same line. Printing `"w\nx\ny\nz"` produces four separate output lines with one character on each line.

11. The remainder operator _____.

a. is represented by a forward slash

b. must follow a division operation

c. provides the quotient of integer division

d. is represented by a percent sign

Answer: d

Feedback:

The remainder operator is the percent sign (%).

12. Which of the following operators has the lowest precedence?

a. multiplication

b. remainder

c. subtraction

d. division

Answer: c

Feedback:

The subtraction operator has lower precedence than the multiplication, division, and remainder operators. The multiplication, remainder, and division operators all have the same precedence.

13. The *equal to* relational operator is _____.

a. =

b. ==

c. !=

d. !!

Answer: b

Feedback:

The `==` operator is the equal to operator. The `=` operator is the assignment operator. The `!=` operator is the *not* equal to operator. The `!!` operator represents two NOT operators.

14. When you perform arithmetic with values of diverse types, Java _____.

a. issues an error message

b. implicitly converts the values to a unifying type

c. requires you to explicitly convert the values to a unifying type

d. implicitly converts the values to the type of the first operand

Answer: b

Feedback:

When you perform arithmetic with values of diverse types, Java converts them to a unifying type. Java does not issue an error message when you perform arithmetic with values of diverse types, it does not require that you explicitly convert the types, and it does not necessarily convert the values to the type of the first operand, but it might if that operand is the unifying type.

15. If you attempt to add a `float`, an `int`, and a `byte`, the result will be a(n) _____.

a. `float`

b. `int`

c. `byte`

d. error message

Answer: a

Feedback:

If you attempt to add a `float`, an `int`, and a `byte`, the result will be a `float`. You will not receive an error message.

16. You use a _____ to explicitly override an implicit type.

a. mistake

b. type cast

c. format

d. type set

Answer: b

Feedback:

You use a type cast to explicitly override an implicit type.

17. In Java, what is the value of 3 + 7 * 4 + 2?

a. 21

b. 33

c. 42

d. 48

Answer: b

Feedback:

Multiplication occurs first, so 3 + 28 + 2 is 33.

18. Which assignment is correct in Java?

a. `int value = (float) 4.5;`

b. `float value = 4 (double);`

c. `double value = 2.12;`

d. `char value = 5c;`

Answer: c

Feedback:

`2.12` can be assigned to a `double`. A `float` cannot be assigned to an `int`. The statement `float value = 4 (double);` does not compile. There is no data type to which `5c` can be assigned.

19. Which assignment is correct in Java?

a. `double money = 12;`

b. `double money = 12.0;`

c. `double money = 12.0d;`

d. All of these are correct.

Answer: d

Feedback:

`12`, `12.0` and `12.0d` can all be assigned to a `double`.

20. Which assignment is correct in Java?

a. `char aChar = 5.5;`

b. `char aChar = "W";`

c. `char aChar = '*';`

d. `char aChar = $;`

Answer: c

Feedback:

The character within single quotes can be assigned to a `char` variable. A numeric value cannot be assigned to a `char` variable. A `String` value cannot be assigned to a `char` variable; the statement with `"W"` would work if the `W` was surrounded by single quotes. A `$` cannot be assigned to a `char` variable; the statement would work if the `$` was surrounded by single quotes.